

**DEPARTMENT OF INFORMATION TECHNOLOGY
NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL.
Assignment 2: Advanced Data Structures - Skip List**

Course Code: IT700

Date: 26.08.2026

Course: Advanced Algorithms

Due Date: 30.08.2025 (by 11:55 PM)

Note:

- *Implement all tasks given below using the Python programming language.*
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1. Write a code implementing self-correcting skip list to maintain **ideal behaviour** showing its list structure.

[CO2- L3, L4, L6]

2. Demonstrate the implementation of a skip list that can be used in routing table Operations.

[CO2- L3, L6]

Course Outcomes:

CO2: Design and Evaluate Different Advanced Data Structures and Their Applications.